

Liquid metal sensors for prosthetic arms



Humans heavily rely on touch receptors. Each human fingertip has more than 3,000 touch receptors. A lack of sensation presents various challenges in carrying out simple tasks such as touching. To fill the absence of these receptors, researchers from Florida Atlantic University's College of Engineering and Computer Science have incorporated stretchable tactile sensors using liquid metal on the fingertips of a prosthetic hand. Encapsulated within silicone based elastomers, this technology provides key advantages over traditional sensors, including high conductivity, compliance, flexibility, and stretchability. This hierarchical multi-finger tactile sensation integration could provide a higher level of intelligence for artificial hands.

Researchers used individual fingertips fitted with stretchable tactile sensors with liquid metal on a prosthesis attached to a robotic arm (Credit: www.fau.edu)

Soft skin patch to monitor blood flow continuously

Engineers at the University of California, San Diego, have developed a soft and stretchy ultrasound patch that can be worn on the skin to monitor blood flow through major arteries and veins deep inside a person's body in real time. Knowing how fast and how much blood flows through the blood vessels can help diagnose various cardiovascular conditions, including blood clots, heart valve problems, poor circulation in the limbs, or artery blockages that could lead to strokes or heart attacks. The patch is made of a thin sheet of flexible, stretchable polymer that adheres to the skin. Embedded on the patch is an array of millimetre-size ultrasound transducers known as an ultrasound phased array. It gives the patch the ability to go deeper and wider.

This soft, stretchy skin patch uses ultrasound to monitor blood flow to organs like the heart and brain (Credit: <https://ucsdnews.ucsd.edu>)



Smartphone screens that can sense soil or water contamination



The touchscreen technology used in billions of smartphones and tablets could also be used as a powerful sensor, without the need for any modifications (Credit: www.cam.ac.uk)

Researchers from the University of Cambridge have demonstrated how a typical touchscreen could be used to identify common ionic contaminants in soil or drinking water by dropping liquid samples on the screen. The researchers pipetted different liquids onto the screen to measure a change in capacitance and recorded the measurements from each droplet using the standard touchscreen testing software. Ions in the fluids interact with the screen's electric fields differently depending on the concentration of ions and their charge. At the moment, the sensitivity of phone and tablet screens is tuned for fingers, but the researchers say the sensitivity could be changed in a certain part of the screen by modifying the electrode design in order to be optimised for sensing.